Robot Vision: Mock Exam 6 (Solutions)

Chapters 2 to 12 · 90 minutes · no documents · 40 points

Part A: Multiple choice

22 points, about 35 min

A1 Which output layer and loss are used for classification into K classes, exactly one being correct? (ch. 2)

(a) K independent sigmoids and squared error.
(b) $K-1$ binary questions with soft labels.
(c) ✓ **K units with softmax and cross-entropy.**
(d) One unit with identity activation.

Softmax makes the K probabilities sum to 1.

A2 Why is reverse-mode automatic differentiation well suited to training neural networks? (ch. 2)

(a) It computes second derivatives for free.
(b) It estimates gradients by finite differences.
(c) ✓ **One backward pass gives the gradient of one scalar loss with respect to all the parameters.**
(d) It gives the derivatives of many outputs with respect to one input.

This is backpropagation.

A3 Which statement about Batch Normalization (BN) and Layer Normalization (LN) is correct? (ch. 3)

(a) BN computes them across the features of each sample; LN across the mini-batch.
(b) ✓ **BN computes statistics across the samples of a mini-batch; LN computes them across the features of each sample.**
(c) LN needs running averages at test time.
(d) Both compute them across the mini-batch.

LN does not depend on the batch: used in Transformers.

A4 What is the main benefit of residual connections in very deep networks? (ch. 3)

(a) ✓ **They give information and gradients a direct path, which makes very deep networks easier to optimize.**
(b) They remove the need for nonlinear activations.
(c) They replace normalization layers.
(d) They reduce the number of parameters.

$z_{l+1} = z_l + F(z_l)$; a useless block can learn $F = 0$.

A5 Which statement about pooling is correct? (ch. 4)

(a) It merges all channels into one map.
(b) ✓ **It combines values in local regions with a fixed rule (max or average), channel by channel, without learned weights.**
(c) It always keeps the spatial resolution.
(d) It uses learned weights to detect patterns.

Pooling never mixes channels.

A6 How does the gradient go back through a max-pooling region with a unique maximum? (ch. 4)

(a) To none of the inputs.
(b) Equally to all inputs of the region.
(c) ✓ **Only to the input that produced the maximum.**
(d) To the smallest input.

Changing a non-maximum input does not change the output.

A7 What characterizes end-to-end learning for steering-angle prediction? (ch. 5)

(a) The network is trained without any data.

(b) It uses hand-designed features.
(c) Lanes are detected first, then a separate controller computes the angle.
(d) ✓ **One network maps camera images directly to steering commands, trained on paired examples.**

NVIDIA's driving network.

A8 How do the skip connections of U-Net recover fine details? (ch. 5)

(a) By adding them, as in ResNet.
(b) ✓ **By concatenating high-resolution encoder features with the upsampled decoder features.**
(c) By pooling the decoder features.
(d) By connecting the input directly to the loss.

U-Net: concatenation; ResNet: addition.

A9 What does a t-SNE plot of the activations of different layers show? (ch. 6)

(a) Distances in t-SNE are exact distances in feature space.
(b) No layer shows any structure.
(c) Lower layers cluster better than higher layers.
(d) ✓ **Higher-layer activations cluster by category more clearly than lower-layer activations.**

t-SNE distances are not faithful.

A10 In the insertion/deletion evaluation, what indicates a good explanation? (ch. 6)

(a) The score drops when salient pixels are inserted.
(b) The score does not change when pixels are deleted.
(c) The map looks like an edge detector.
(d) ✓ **The score rises quickly when the most salient pixels are inserted and drops quickly when they are deleted.**

No single test proves faithfulness.

A11 Why can horizontal flipping be a bad augmentation for digit recognition but fine for cats? (ch. 7)

(a) Flipping changes the image size.
(b) Cats are symmetric and digits are not black and white.
(c) Flipping is always forbidden for small images.
(d) ✓ **A flipped digit may no longer be a valid digit, so its label would be wrong.**

Augmentation must keep a valid target.

A12 How does a prototypical network classify a query image? (ch. 7)

(a) ✓ **By its distance to each class prototype, the mean feature of that class's support samples.**
(b) By nearest neighbor on raw pixels.
(c) By a softmax trained on all base classes.
(d) By the label of the largest support set.

Few-shot learning.

A13 Which situation illustrates epistemic uncertainty? (ch. 8)

(a) The label of the image is noisy.
(b) The image is so blurry that two objects cannot be told apart.
(c) ✓ **The model has seen too few examples of this kind of object appearance.**
(d) The sensor adds random noise.

Epistemic is reducible with more data; the others are aleatoric.

A14 Why is a temperature $T > 1$ used in knowledge distillation? (ch. 8)

(a) ✓ **To soften the outputs so the small probabilities between classes become visible to the student.**
(b) To change the predicted class of the teacher.
(c) To speed up the teacher.

(d) To make the teacher's outputs one-hot.

Same T for teacher and student.

A15 The points of a point cloud are reordered without changing their coordinates. How should the outputs behave? (ch. 9)

(a) ✓ **Classification stays the same; per-point segmentation is reordered in the same way.**
(b) Both outputs stay exactly the same.
(c) Both outputs change unpredictably.
(d) Classification is reordered; segmentation stays the same.

Invariance vs equivariance.

A16 What do the attentional GNN and the matching layer of SuperGlue do? (ch. 9)

(a) The GNN detects keypoints; the matching layer uses NMS.
(b) Both only work on one image.
(c) The matching layer is trained separately with labels.
(d) ✓ **The GNN exchanges information within and across the two keypoint sets; the matching layer uses optimal transport with a dustbin for unmatched points.**

Sinkhorn; trained end to end.

A17 Why does the Transformer decoder use masked self-attention? (ch. 10)

(a) To make the cost linear.
(b) To replace positional encoding.
(c) ✓ **So that a position cannot attend to the future tokens it must predict.**
(d) To attend to the encoder only.

No peeking during training.

A18 What lets DETR detect objects without NMS? (ch. 10)

(a) A CNN head that removes duplicates.
(b) ✓ **Detection as set prediction with one-to-one bipartite matching between predictions and objects during training.**
(c) Predicting one box per image.
(d) Anchors with a high IoU threshold.

Hungarian matching; object queries.

A19 Which architecture dominates today's generative LLMs? (ch. 11)

(a) Encoder-only Transformers such as BERT.
(b) Recurrent networks such as LSTMs.
(c) Convolutional networks.
(d) ✓ **Decoder-only Transformers such as GPT.**

BERT: context from both sides, for classification and features.

A20 In LLaVA, what connects the vision encoder to the LLM? (ch. 11)

(a) ✓ **A learned projection that maps the visual features into the LLM's input embedding space.**
(b) A new LLM trained from scratch.
(c) Nothing: the LLM reads pixels directly.
(d) A fixed, hand-designed mapping.

Then visual instruction tuning.

A21 How does CycleGAN learn without paired images? (ch. 12)

(a) By maximizing the exact likelihood.
(b) By using a single mapping and an L1 loss.
(c) ✓ **With two mappings $G : X \to Y$ and $F : Y \to X$ and a cycle consistency loss $F(G(x)) \approx x$.**
(d) By pairing images by hand.

Plus adversarial losses in each domain.

A22 In classifier-free guidance, what happens with a guidance scale $s > 1$? (ch. 12)

(a) The model ignores the prompt.
(b) The denoiser must be retrained.
(c) **✓ The prediction is pushed further toward the condition: better prompt adherence, but less diversity and possible artifacts.**
(d) Sampling becomes deterministic and exact.

$\hat{\varepsilon} = \varepsilon(\varnothing) + s[\varepsilon(c) - \varepsilon(\varnothing)]$.

Part B: Short answers

12 points, 3 per question, about 35 min

B1 What does CAM require, and how does Grad-CAM remove this requirement? (ch. 6)

CAM needs global average pooling followed by a linear classifier, whose weights weight the feature maps. Grad-CAM uses the averaged gradients of the class score as weights, so it works on any CNN.

B2 Why is face recognition treated as metric learning rather than classification? (ch. 8)

The people seen at test time may not be in the training set. The network learns a space where distance means identity, and a query is compared with gallery features.

B3 Why does DETR not need non-maximum suppression? (ch. 10)

Detection is treated as set prediction: during training, bipartite (Hungarian) matching assigns each object to exactly one query, so the model learns not to produce duplicates.

B4 Compare pix2pix and CycleGAN: what data does each need? (ch. 12)

pix2pix needs paired images (input and desired output), and its discriminator sees the pair. CycleGAN works with unpaired collections, using two mappings and a cycle consistency loss.

Part C: Calculations

6 points, 2 per question, about 15 min

C1 Dropout with $p = 0.5$ is used during training. By how much are the surviving activations scaled? (ch. 3)

$1/(1-p) = \mathbf{2}$.

C2 In a GNN, a node must receive information from a node 3 edges away. How many message-passing layers are needed at least? (ch. 9)

One layer = one hop, so at least **3 layers**.

C3 Classifier-free guidance: $\varepsilon(\varnothing) = 0.1$, $\varepsilon(c) = 0.3$. Compute $\hat{\varepsilon}$ for $s = 0$, 1 and 2. (ch. 12)

$\hat{\varepsilon} = \varepsilon(\varnothing) + s[\varepsilon(c) - \varepsilon(\varnothing)]$: $s = 0$: **0.1**; $s = 1$: **0.3**; $s = 2$: $0.1 + 2 \times 0.2 = \mathbf{0.5}$.